Ch 8: Heredity

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SECTION A - INTRODUCTION TO HEREDITY AND VARIATION

1. Meaning of heredity

- Heredity is the transmission of characteristics (traits) from parents to their offspring through genes.
- It ensures continuity of life from one generation to the next.

2. Meaning of variation

- Variation refers to the differences in characteristics among individuals of the same species.
- It arises due to differences in genes and their combinations, and also due to environmental influences.

3. Why offspring resemble parents but are not exactly identical

- Offspring inherit genes from both parents. Half the genes come from the father and half from the mother.
- Each parent has two versions (alleles) of a gene. The offspring receives one allele for each gene.
- Random combination of alleles during gamete formation and fertilisation creates new combinations.
- Small differences in genes or in the environment lead to visible variations.

4. Importance of variation in survival and adaptation

- Variation helps a species survive in changing environments.
- Some individuals with favourable variations can adapt better and survive, while others may not.
- These survivors can pass on their useful traits to the next generation.
- Variation is the raw material for evolution.

5. Inherited traits vs Acquired traits

- Inherited traits are present from birth and are passed on by genes from parents.
- Acquired traits develop during life due to environment, practice, diet, or training and are not passed to offspring.

6. Basic relationship: Cell → Nucleus → Chromosomes → Genes → Traits

- All living organisms are made of cells.
- The nucleus of a cell contains chromosomes.
- Chromosomes are made of DNA and proteins.
- Genes are specific segments of DNA on chromosomes.
- Genes carry information that determines traits.

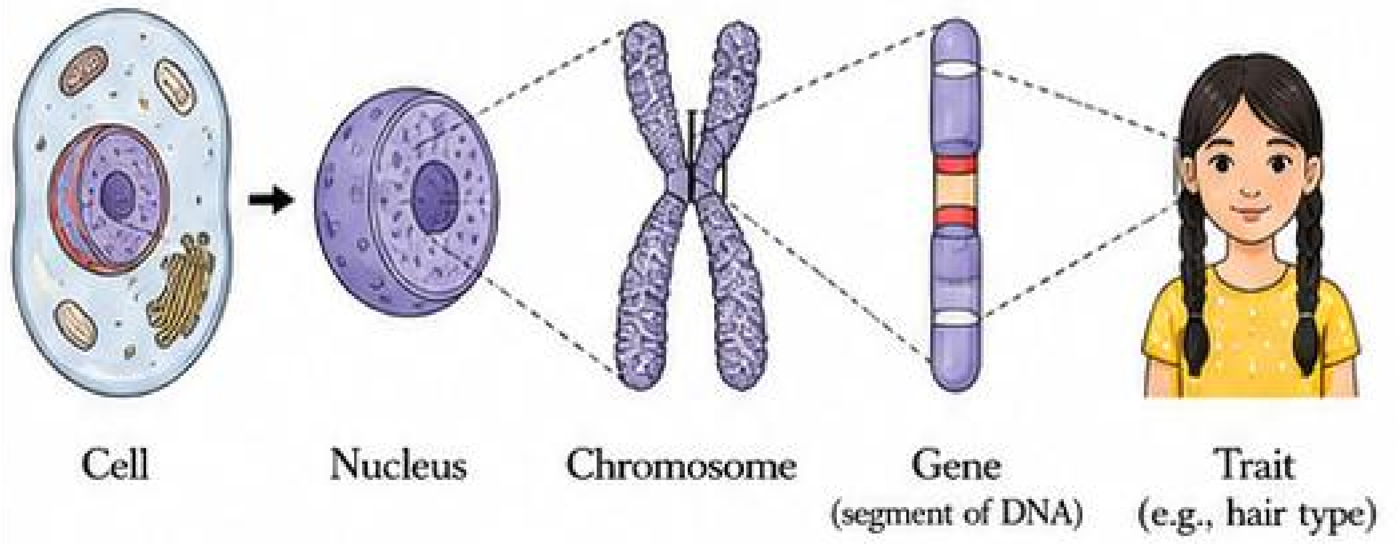
7. DNA as hereditary material

- DNA (Deoxyribonucleic Acid) is the molecule that stores genetic information.
- It is present in the chromosomes of the nucleus.
- DNA can copy itself during cell division.
- The copied DNA is passed to daughter cells and to the next generation during reproduction.
- Watson and Crick discovered the double helix structure of DNA in 1953.

8. Everyday examples of heredity and variation

- Resemblance: Children may have their father's eye colour or mother's type of hair.
- Variation: In a family, children may have different heights, skin tones, or hair types.
- Siblings show similarities but are not exactly the same.
- Example in animals: Puppies in the same litter may differ in colour, size, or markings.

Cell → Nucleus → Chromosome → Gene → Trait



Flow Summary

Cells contain a nucleus → Nucleus contains chromosomes → Chromosomes are made of DNA → Genes are segments of DNA → Genes control traits.

Heredity vs Variation

Heredity	Variation
Transmission of traits from parents to offspring.	Differences in traits among individuals of the same species.
Ensures continuity of species.	Creates diversity within a species.
Based on genetic similarity.	Caused by differences in genes and environment.
Traits are passed from one generation to the next.	Variation helps in adaptation and survival.
Example: Eye colour, blood group are inherited.	Example: Height, skin colour vary among individuals.

Inherited Traits vs Acquired Traits

Feature	Inherited Traits	Acquired Traits
Origin	Passed from parents through genes.	Develop during life due to environment, practice, diet, or training.
Present from birth?	Yes	No, develop after birth.
Examples	Eye colour, ear lobe type, blood group, hair type, height potential.	Tanned skin (from sun), body building, learning to ride a cycle.
Passes to offspring?	Yes	No
Controlled by	Genes	Environment and personal effort

KEY IDEA

Traits are passed through genes, but variations arise because copying is not perfectly identical.

EXAM TIPS – CENTUM FOCUS

1. Heredity transmits traits; variation creates differences.
2. Genes on chromosomes carry information for traits.
3. Offspring inherit genes from both parents.
4. Inherited traits are genetic; acquired traits are not passed on.
5. Variation is essential for adaptation and evolution.

QUICK RECALL

- Heredity = Transmission of traits
- Variation = Differences in traits
- DNA carries genetic information
- Genes control traits
- Variation helps in survival

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SECTION B - GENES, DNA, CHROMOSOMES AND EXPRESSION OF TRAITS

1. Important terms and their meanings

- **Gene:** A unit of heredity. It is a segment of DNA that codes for a protein and controls a trait.
- **Allele:** Different forms of the same gene that control the same trait.
- **DNA (Deoxyribonucleic Acid):** The chemical that carries hereditary information in cells.
- **Chromosome:** Thread-like structures made of DNA and proteins. They carry many genes.
- **Nucleus:** A small, round structure inside a cell that contains chromosomes.

2. From DNA to trait: the flow of information

- DNA contains genes. A gene makes a protein. Proteins do the work in cells and control the structure and function of the body. This produces a visible trait.

DNA	→	Gene	→	Protein	→	Trait
(Information)		(Code)		(Product)		(Visible result)

3. Chromosomes occur in pairs

- In body (somatic) cells, chromosomes occur in pairs. One chromosome of each pair comes from the father and the other from the mother.
- Human body cells have 46 chromosomes arranged in 23 pairs.
- Sex cells (sperm and egg) have half the number (23 chromosomes).

4. Alleles: different forms of a gene

- A single gene can exist in different forms called alleles.
- Example: Gene for height may have allele *T* (tall) and allele *t* (short).

5. Genotype and Phenotype

- **Genotype:** The genetic makeup of an individual. It is the combination of alleles an individual has for a trait.
- **Phenotype:** The visible appearance or expression of a trait.
- Example: Genotype TT or Tt → Phenotype Tall

6. Homozygous and Heterozygous

- **Homozygous:** When both alleles for a gene are the same. Example: TT or tt
- **Heterozygous:** When the two alleles for a gene are different. Example: Tt

7. How variations arise during reproduction

- During reproduction, DNA is copied to pass genetic information to the next generation.
- DNA copying is not 100% exact. Small, random changes (mutations) can occur.
- These changes create new alleles or traits. Recombination of genes during gamete formation also produces new combinations.

8. Why proteins control body design and traits

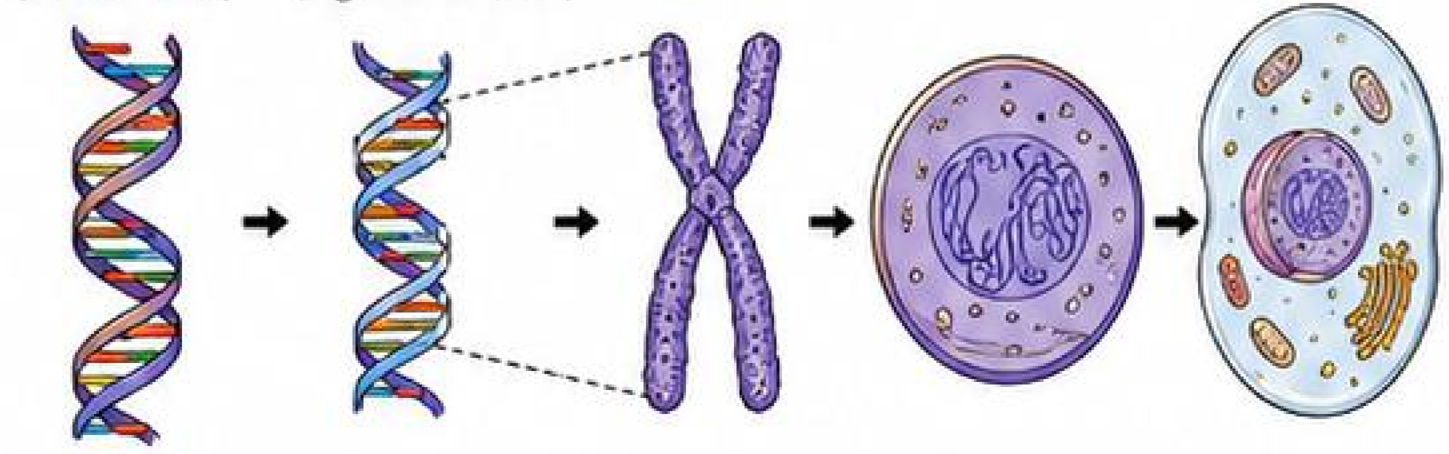
- Proteins are made of amino acids and have specific shapes.
- They act as enzymes, hormones, structural parts and transporters in the body.
- The type and amount of proteins produced determine how a trait appears. Hence, proteins control body design and functions.

KEY IDEA

Genes carry the code.
Proteins do the work.
The result is the trait we see.

Small changes in DNA and new combinations of alleles create variations in the next generation.

The journey of hereditary information



Explanation

A small part of DNA is a gene. Many genes are packed in chromosomes. Chromosomes are present in the nucleus. The nucleus is found in every cell. This information ultimately produces proteins and shows as traits.

Key terms at a glance

Term	What it is	Key points
Gene	Segment of DNA	Codes for a protein; controls a trait.
Allele	Different form of the same gene	Control the same trait in different ways (e.g., T or *t* for height).
DNA	Molecule in nucleus	Carries genetic information; arranged as a double helix.
Chromosome	Structure of DNA + proteins	Carries many genes; occurs in pairs in body cells.
Trait	Visible or measurable characteristic	Result of proteins produced from genes (e.g., eye colour).

Important comparisons

Genotype vs Phenotype

Genotype	Phenotype
• Genetic makeup (allele combination). • Cannot be seen. • Example: Tt	• Physical expression of the trait. • Can be seen or measured. • Example: Tall

Homozygous vs Heterozygous

Homozygous	Heterozygous
• Two identical alleles. • Example: TT or tt • Pure form.	• Two different alleles. • Example: Tt • Hybrid form.

Simple everyday examples

- **Eye colour:** Gene for eye colour has different alleles (e.g., Brown 'B' and Blue 'b').
- **Tongue rolling:** Some people can roll their tongue (allele R), others cannot (allele *r*).
- **Blood group:** Gene for blood group has alleles I^A, I^B and *i*.
- **Height:** Many genes and environment together decide whether a person is tall or short.

EXAM TIPS – CENTUM FOCUS

1. Remember the flow: DNA → Gene → Protein → Trait.
2. Learn definitions of gene, allele, DNA, chromosome, nucleus.
3. Distinguish between genotype and phenotype clearly.
4. Use examples (eye colour, tongue rolling, blood group, height) in answers.
5. Explain how variations arise during reproduction.

QUICK RECALL

- Gene – unit of heredity.
- Allele – different forms of a gene.
- DNA – carries genetic information.
- Chromosome – thread-like, in pairs.
- Nucleus – contains chromosomes.
- DNA → Gene → Protein → Trait.
- Homozygous: same alleles.
- Heterozygous: different alleles.

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SECTION C - GREGOR MENDEL AND THE BEGINNING OF GENETICS

1. Who was Gregor Mendel?

- Gregor Johann Mendel (1822–1884) was an Austrian monk and a scientist.
- He conducted experiments on pea plants in the monastery garden of St. Thomas in Brünn (now Brno, Czech Republic).
- He discovered the basic principles of heredity by studying how traits are passed from parents to offspring.
- His work was published in 1866 but was not recognized widely until the early 20th century.
- He is called the "Father of Genetics" because he was the first to scientifically explain inheritance through experiments and mathematical analysis.

2. Why did Mendel choose pea plants?

- **Short life cycle:** A pea plant completes its life cycle in about 3–4 months.
- **Many contrasting (distinct) traits:** Easy to observe and record.
- **Easy control of pollination:**
  - Plants are naturally self-pollinated.
  - Cross-pollination can be done easily by hand.
- **Many seeds:** A large number of seeds in each pod allow accurate results.
- Grows well in available space and requires simple care.

3. Important Terms

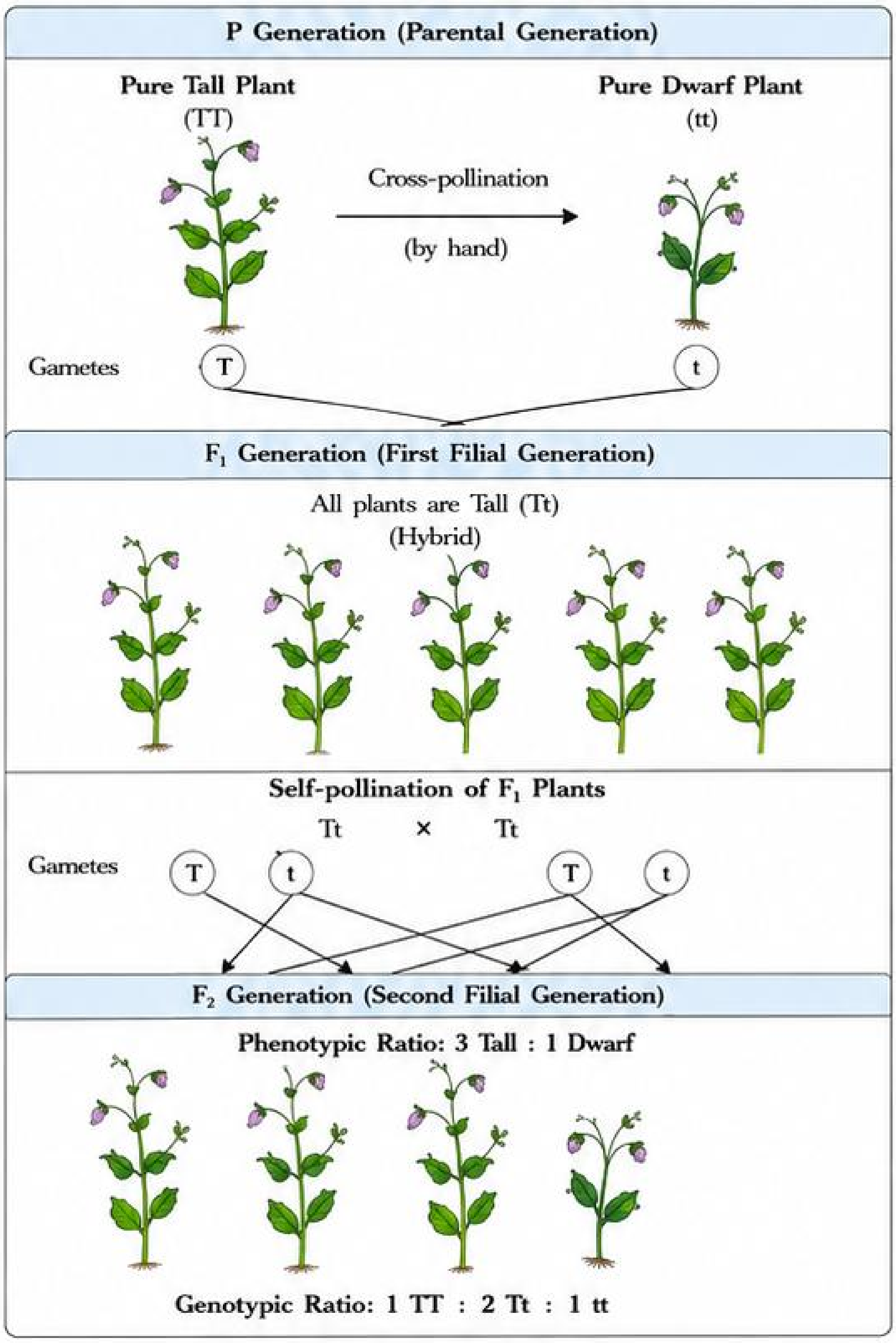
- **Pollination:** Transfer of pollen from the anther (male part) to the stigma (female part) of a flower.
- **Self-pollination:** Pollen is transferred within the same flower or between flowers of the same plant.
- **Cross-pollination:** Pollen is transferred from the flower of one plant to the flower of another plant of the same species.
- **Pure line:** A plant that, when self-pollinated for several generations, produces offspring with the same trait. (Homozygous condition)

4. Important Contrasting Traits in Pea Plants

S. No.	Trait	Dominant Form	Recessive Form
1.	Plant height	Tall	Dwarf
2.	Seed shape	Round	Wrinkled
3.	Seed colour	Yellow	Green
4.	Flower colour	Violet	White
5.	Pod shape	Inflated	Constricted
6.	Pod colour	Green	Yellow
7.	Flower position	Axial	Terminal

5. Mendel's Monohybrid Experiment (Plant Height)

Mendel studied one trait at a time. Let us see his experiment on plant height.

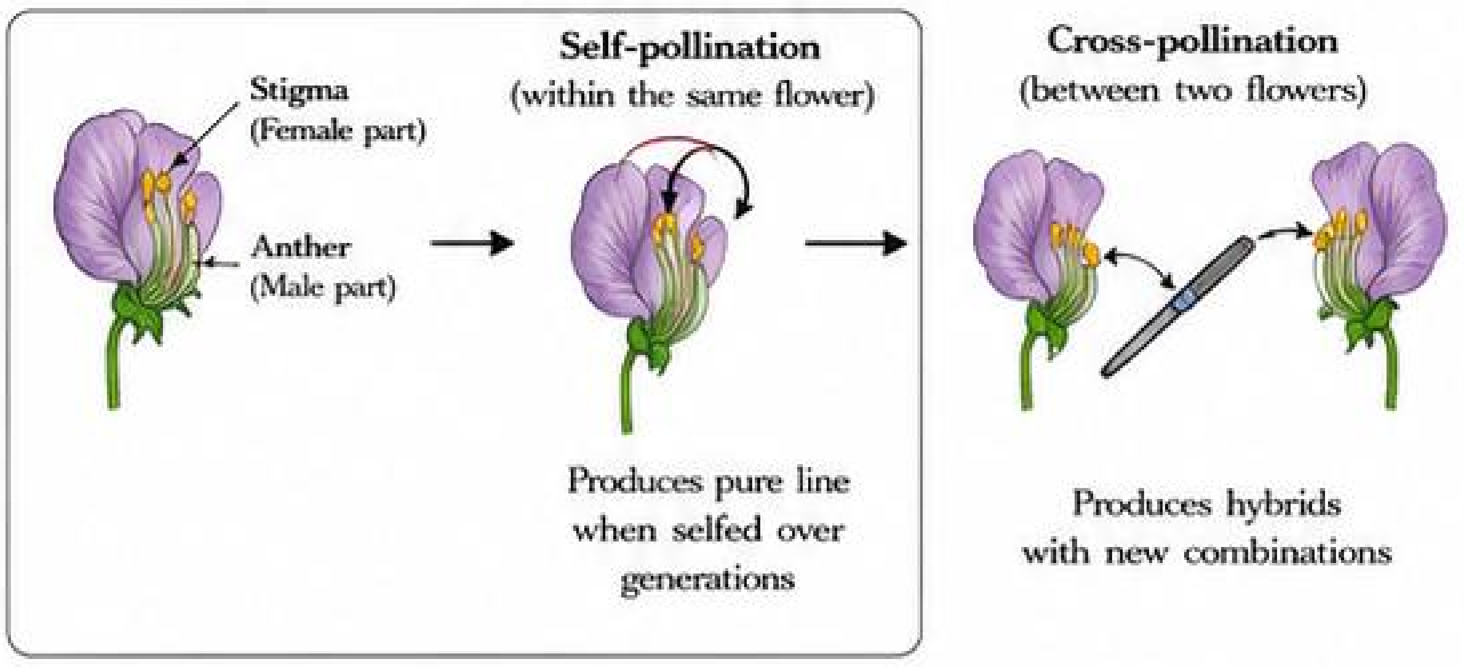


6. Dominant and Recessive Traits

- When two contrasting traits are present together, only one trait is expressed, and the other is masked.
- The expressed trait is called **Dominant**.
- The masked trait is called **Recessive**.
- The dominant trait shows up even in the presence of the recessive trait.
- The recessive trait appears only when both alleles for that trait are recessive.

Feature	Dominant Trait	Recessive Trait
Expression	Expressed even in hybrid condition	Masked in hybrid condition
Allele	Represented by a capital letter (e.g., T)	Represented by a small letter (e.g., t)
Example	Tall (T), Yellow seed (Y)	Dwarf (t), Green seed (y)
Combination	TT or Tt show dominant trait	tt shows recessive trait

7. Pea Flower and Pollination (Diagram)



KEY TAKEAWAY

Mendel's careful experiments on pea plants led to the discovery of the basic laws of inheritance—Law of Dominance, Law of Segregation and Law of Independent Assortment.

EXAM TIPS – CENTUM FOCUS

1. Remember: Mendel → Pea plant → 7 contrasting traits.
2. For monohybrid cross: F_1 all dominant; F_2 ratio 3:1.
3. Capital letter = Dominant; small letter = Recessive.
4. Pure line = same trait on selfing for many generations.
5. Diagrams and ratios are frequently asked in exams.

QUICK RECALL

- Pea plant chosen for: short life cycle, contrasting traits, easy pollination control, many seeds.
- Pollination: transfer of pollen.
- Self-pollination: same plant.
- Cross-pollination: different plant.
- Pure line: true breeding.
- Dominant masks, recessive hidden.

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SECTION D - MONOHYBRID CROSS IN DETAIL

1. Mendel's tall/dwarf pea experiment

- Mendel chose pea plants because they are easy to grow, have many varieties and show clear contrasting traits.
- He selected the trait 'height of stem': tall and dwarf.
- He performed controlled crosses and observed traits over two generations.

2. Symbols for the trait

- **T** = tall (dominant allele)
- t = dwarf (recessive allele)
- In pea plants, tall (T) is dominant over dwarf (t).

3. Meaning of terms

Pure tall (Homozygous dominant)	TT – both alleles are T. Always tall. Gives only T gametes.
Pure dwarf (Homozygous recessive)	tt – both alleles are t. Always dwarf. Gives only t gametes.
F_1 hybrid (Heterozygous)	Tt – one T and one t. Looks tall (because T is dominant). Gives both T and t gametes.

4. Parental cross (P generation)

Mendel crossed a pure tall plant with a pure dwarf plant.

Pure tall TT × Pure dwarf tt → All F_1 plants are tall (Tt)

5. F_1 selfing (Tt × Tt)

- Mendel allowed F_1 plants (Tt) to self-fertilize.
- Each F_1 plant produces two types of gametes: T and t.
- The combination of gametes is shown below.

Punnett Square for Tt × Tt

Female Gametes ↓ / Male Gametes →	T	t
T	TT (Tall)	Tt (Tall)
t	Tt (Tall)	tt (Dwarf)

6. Results in F_2 generation

- **Genotype ratio:** 1 TT : 2 Tt : 1 tt
- **Phenotype ratio:** 3 Tall : 1 Dwarf

Genotype	1 TT	2 Tt	1 tt
Phenotype	3 Tall		1 Dwarf

7. Why is this so?

- All F_1 plants are tall (Tt) because the dominant allele T masks the effect of recessive allele t. This is called the *law of dominance*.
- In F_2 generation, when F_1 plants self, the recessive alleles (t) come together (tt) in some offspring and dwarf plants appear.
- This shows that the recessive trait is not lost; it remains hidden in F_1 and reappears in F_2.

8. Law of Dominance

When two different alleles of a gene are present together in a hybrid, only the dominant allele expresses itself, while the recessive allele remains masked.

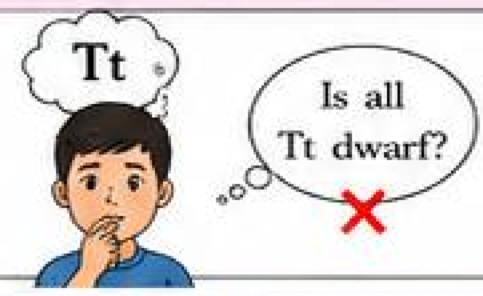
Example: In Tt, tall (T) is expressed; dwarf (t) is masked.

9. Law of Segregation

The two alleles of a gene separate (segregate) during gamete formation so that each gamete receives only one allele.
These alleles unite randomly at fertilization.
This law explains the 1:2:1 genotype ratio in F_2.

10. Common mistakes students make (Genotype vs Phenotype)

- Thinking genotype and phenotype are the same. → Genotype is genetic makeup (TT, Tt, tt); phenotype is the visible trait (tall or dwarf).
- Saying all Tt are different. → All Tt look tall; only tt are dwarf.
- Confusing ratios. → 1:2:1 is genotype ratio; 3:1 is phenotype ratio.
- Forgetting dominance. → Recessive trait appears only in tt, not in Tt.
- Writing gametes incorrectly. → Tt produces T and t, not Tt.



11. Solved examples

Example 1: A tall pea plant (pure) is crossed with a dwarf pea plant (pure). What will be the phenotype of F_1 and the ratio in F_2?

Solution:
P generation: TT × tt
F_1 generation: All Tt (100% tall)
F_1 selfing: Tt × Tt
F_2 genotype ratio = 1 **TT** : 2 Tt : 1 tt
F_2 phenotype ratio = 3 Tall : 1 Dwarf
Answer: F_1 – all tall; F_2 – 3 tall : 1 dwarf.

Tall (TT)

×

Dwarf (tt)

Example 2: Two F_1 hybrid (Tt) pea plants are crossed. What are the chances of getting a dwarf plant in F_2?

Solution:
Tt × Tt (use Punnett square)
Dwarf plants are tt.
Out of 4 boxes, 1 box is tt.
Chance of dwarf = $\frac{1}{4}$ = 25%

Answer: 25% chance of dwarf plants.

12. Practice corner

Q1. A tall pea plant (TT) is crossed with a dwarf pea plant (tt). What will be the phenotype of the F_1 generation?
Ans: All plants will be tall (Tt).

Q2. Two F_1 hybrid plants (Tt) are crossed. What are the genotype and phenotype ratios in F_2?
Ans: Genotype ratio = 1 **TT** : 2 Tt : 1 tt
Phenotype ratio = 3 Tall : 1 Dwarf

KEY IDEA

- Dominant trait expresses in F_1.

Recessive trait hides in F_1 but reappears in F_2.

Genotype explains makeup; phenotype shows appearance.

EXAM TIPS – CENTUM FOCUS

1. Always write symbols clearly: T (tall) and t (dwarf).
2. Draw a neat Punnett square and fill all boxes.
3. State both genotype and phenotype ratios.
4. Use key terms: dominant, recessive, homozygous, heterozygous, segregation.
5. If a plant is dwarf, its genotype must be tt.

QUICK RECALL

- T = Tall (dominant)
- t = Dwarf (recessive)
- Pure tall = TT
- Pure dwarf = tt
- F_1 hybrid = Tt (appears tall)
- F_1 selfing: Tt × Tt
- Genotype ratio = 1:2:1
- Phenotype ratio = 3:1

CENTUM FOCUS

Think like Mendel!

- Write symbols correctly.
- Use Punnett square.
- Don't confuse 1:2:1 with 3:1.
- Practice more monohybrid crosses to score full marks.

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SECTION E - HOW VARIATIONS ARISE AND HOW TRAITS ARE INHERITED

1. Sexual reproduction mixes genes from two parents
- In sexual reproduction, offspring are produced by the union of gametes (sex cells) from two parents.
- Each parent contributes half the genetic material in the form of gametes.
- During fertilisation, the gametes fuse and form a zygote with a new combination of genes.

2. Gamete formation halves the chromosome number
- In the reproductive organs, special cell division called meiosis produces gametes.
- Meiosis reduces the chromosome number to half.
- This ensures that when two gametes fuse, the normal chromosome number of the species is restored.

3. Fertilisation restores the number and creates new combinations
- When male and female gametes fuse during fertilisation, the chromosome number is restored.
- The genes from both parents combine in new ways.
- Each child therefore inherits a unique combination of traits, different from both parents and siblings.

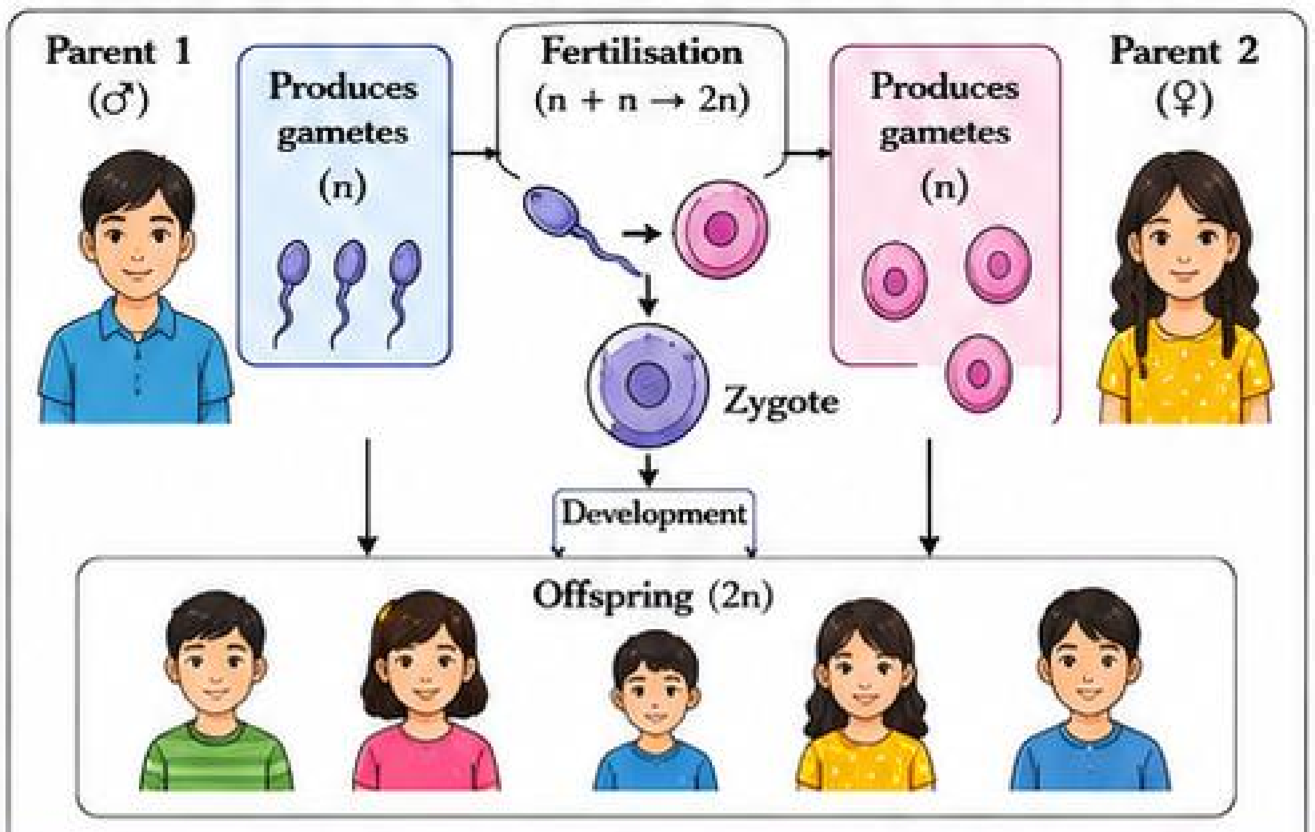
4. DNA copying is reliable but not absolutely perfect
- Before cells divide, DNA is copied so that genetic information is passed on.
- The copying process is highly accurate but small mistakes can occur.
- These changes are called mutations.
- Mutations introduce new genes or new versions of genes, adding to variation.

5. Simple sequence of inherited trait transmission

Parent traits (known genes) → Gametes (half set) → Fertilisation (fusion) → Offspring traits (new combination)

- Traits are passed from parents to offspring through genes.
- Offspring show a mix of traits from both parents and also new combinations.

6. Flow diagram: How reproduction creates variation



New combinations of genes + possible mutations = Variation among offspring

Sexual Reproduction vs. Environmental Effects

Variation due to Sexual Reproduction (Inherited)	Changes due to Environment (Not Inherited)
Caused by mixing of genes and mutations during reproduction.	Caused by external factors like diet, climate, training, habits.
Present from birth.	Develop during life.
Passed from parents to offspring through genes.	Not passed to offspring.
Permanent (inherited in body cells and germ cells).	Temporary or limited to the individual.
Examples: eye colour, blood group, hair type, height potential.	Examples: body building, scars, language, pierced ears, tanned skin.

Examples: Inherited vs. Acquired Traits

Inherited Traits	Acquired Traits
Eye colour	Body building (muscle development)
Blood group	Scars from injuries
Hair type (straight/curly)	Language learned (e.g., English, Hindi)
Height potential	Pierced ears
Earlobe type (attached/free)	Tanned skin from sunlight
Taste preference (in some cases)	Calluses from hard work

Sources of Variation

Source	How it Creates Variation	Example
Recombination of genes (independent assortment)	During meiosis, chromosomes assort independently, creating new combinations in gametes.	Different combination of height and skin colour genes.
Crossing over	Exchange of DNA between homologous chromosomes during meiosis.	New gene combinations on chromosomes.
Random fertilisation	Any one of millions of sperm can fuse with any one of millions of eggs.	Unique gene mix in each child.
Mutation	Small changes in DNA sequence create new genes or gene versions.	Extra finger, different hair texture, colour blindness.
Gene flow	Movement of genes from one population to another through migration.	New traits introduced in a population.

Why is Variation Useful?
- ✓ Variation increases the chances that some individuals will survive changes in the environment.
- ✓ It helps populations adapt to new conditions.
- ✓ It is the raw material for evolution.
- ✓ Without variation, all members would be alike and a single change could wipe out the whole population.

KEY IDEAS
- Sexual reproduction mixes genes from two parents.
- Gametes have half the chromosomes.
- Fertilisation restores the number and creates new combinations.
- DNA copying errors cause mutations, adding variation.

EXAM TIPS – CENTUM FOCUS
1. Always link variation to reproduction, meiosis and mutations.
2. Remember: meiosis → half number; fertilisation → full number.
3. Distinguish inherited (genetic) vs. acquired (environmental) traits.
4. Use examples: eye colour, blood group (inherited); scars, language (acquired).
5. Diagrams and flowcharts score high in board exams.

QUICK RECALL
- Mix of genes + mutation = Variation.
- Inherited traits: passed through genes, present from birth.
- Acquired traits: due to environment, not passed on.
- Variation helps in survival and adaptation.

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SECTION F - INHERITED TRAITS VS ACQUIRED TRAITS (DEEP STUDY)

1. What are Inherited Traits?

- Inherited traits are characteristics that are passed from parents to offspring through genes.
- They are determined by the DNA present in the reproductive cells (gametes) – sperm and ovum.
- These traits are present from birth (or develop early) and remain throughout life.
- They are not usually affected by day-to-day environment.

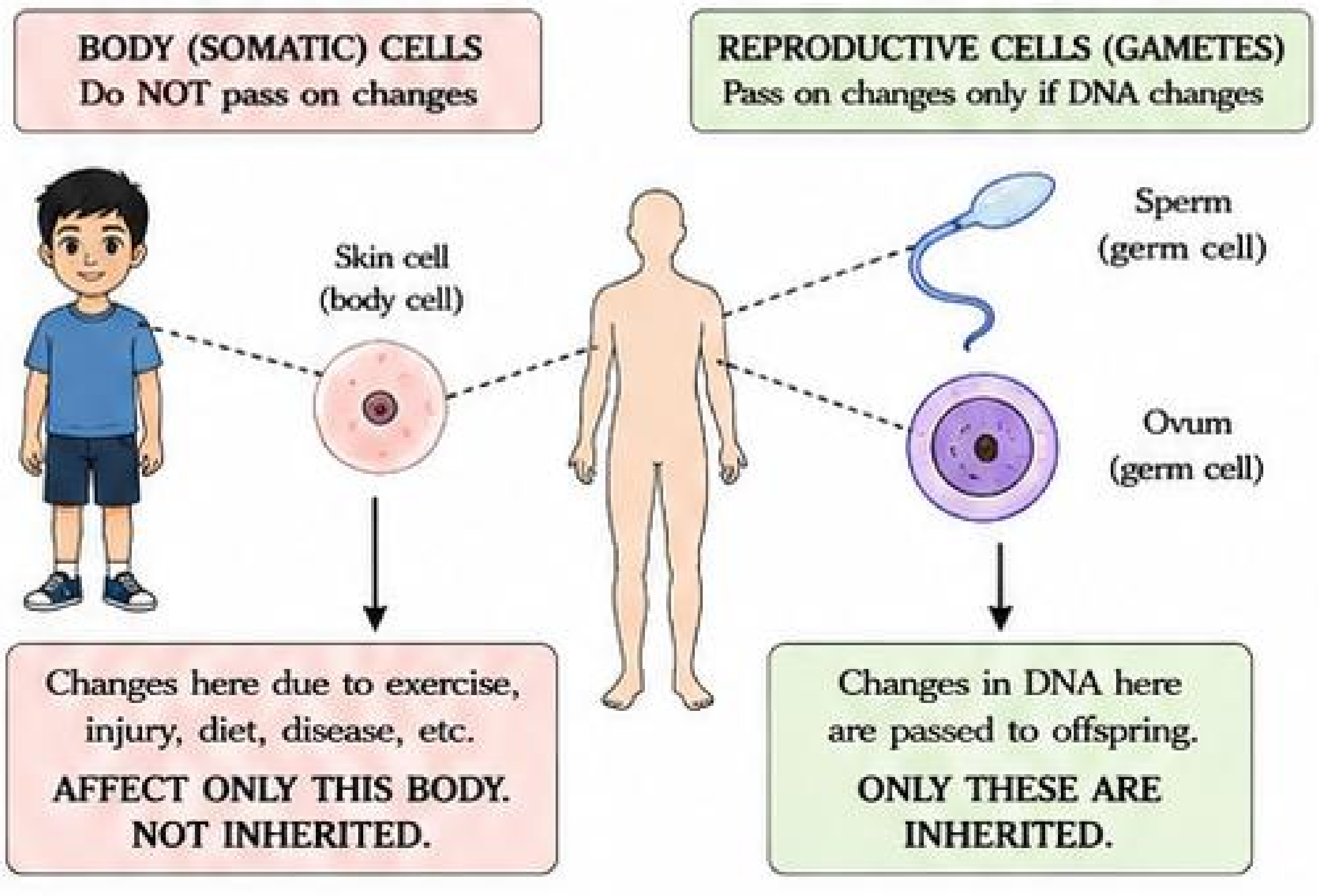
2. What are Acquired Traits?

- Acquired traits are characteristics developed by an individual during its lifetime due to environment, habits, diet, exercise, training, or disease.
- They are not present at birth.
- They affect only the body (somatic) cells and not the genes in the reproductive cells.
- They are not passed to the next generation.

3. Why Acquired Traits are NOT Passed to Offspring?

- For a trait to be inherited, there must be a **change in the DNA** of the **reproductive cells (gametes).**
- Acquired traits affect only the body (somatic) cells.
- These changes do not reach the germ cells.
- During reproduction, only germ cells (with original DNA) unite to form the new individual.
- Therefore, acquired traits are not passed on.
- **Only changes in DNA of germ cells can be inherited.**

Body Cells vs Reproductive Cells – What Gets Inherited?



Only changes in the DNA of germ cells (gametes) are inherited.

Inherited Traits vs Acquired Traits – Comparison

Feature	Inherited Traits	Acquired Traits
Origin	• Present in genes (DNA). • Passed from parents to offspring.	• Developed during life due to environment, habits, training, diet, injuries, diseases, etc.
Control	• Controlled by genes.	• Controlled by environment and personal efforts.
Permanence	• Usually constant throughout life. • May develop at different stages but remain stable.	• May change with conditions. • Can be lost if cause is removed.
Transmission to Offspring	• Passed from one generation to the next through reproductive cells (gametes). • Present from birth or develop early.	• Not passed to offspring. • Affects only the individual.
Basis	• Change in DNA of germ cells.	• Change in body (somatic) cells only.
Examples (Humans)	• Eye colour, blood group, ear lobe type, hair type, height potential, sex, skin colour.	• Tanned skin, body building, shaved head, learned skills (cycling, piano), pierced ears.
Examples (Animals)	• Coat colour in dogs, horns in cattle, beak shape in birds, tail length in cats.	• Trained behaviour (circus tricks), muscle strength, injury scars.
Examples (Plants)	• Flower colour in pea, seed shape in bean, leaf shape, plant height potential.	• Size of plant due to more water/fertiliser, leaf size in shade, grouped roots due to soil compaction.

Many Examples – Inherited vs Acquired

Inherited Traits (Born with or appear early)

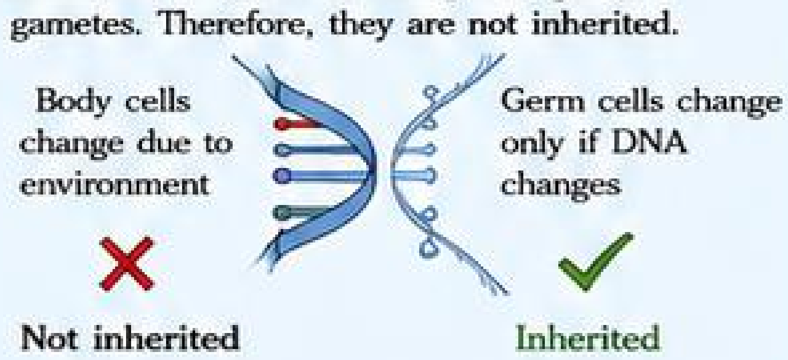
- **Human:** Eye colour, blood group, sex, curly/straight hair, earlobe (attached/free).
- **Animal:** Coat colour in cats, horns in deer, beak shape in finches.
- **Plant:** Flower colour in pea, seed shape in bean, leaf shape in hibiscus, height potential in maize.

Acquired Traits (Developed during life)

- **Human:** Body building, tanned skin, calluses on hands, learned skills, pierced nose.
- **Animal:** Trained dog to sit, pigeons trained to return home, scar after injury.
- **Plant:** Bigger leaves in shade, stunted growth due to lack of light, thicker leaves in sun.

Key Point

Acquired traits may help an individual survive better, but they do not change the genes in gametes. Therefore, they are not inherited.



Short Answered Questions (Exam-Oriented)

1. **Define inherited traits.**
   Ans: Inherited traits are characteristics passed from parents to offspring through genes present in reproductive cells.
2. **Define acquired traits.**
   Ans: Acquired traits are characteristics developed during life due to environment, habits, training, diet, etc., and are not passed to offspring.
3. **Can the child of a bodybuilder be born muscular?**
   Ans: No. Muscular body is an acquired trait due to exercise. It affects body cells only, not the genes in gametes.
4. **Why does a plant grown in shade not pass that exact change to offspring?**
   Ans: Shade causes temporary changes in leaf size and colour in body cells only. Genes in gametes remain unchanged, so offspring do not show the same change.
5. **What is the only way an acquired trait can become inherited?**
   Ans: If there is a change in the DNA of germ cells (mutations), it can be inherited. Normal acquired traits do not cause such changes.

EXAM TIPS – CENTUM FOCUS

1. Remember: Inherited = gene based = passed on.
2. Remember: Acquired = environment based = not passed on.
3. If the change affects only body cells → Not inherited.
4. If the change is in germ cells (DNA) → Inherited.
5. Use clear examples from humans, animals and plants in answers.

QUICK RECALL

- Inherited traits come from genes.
- Acquired traits come from environment.
- Only DNA change in germ cells is inherited.
- Acquired traits benefit the individual, not the next generation.

Ch 8: Heredity

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SECTION G - SEX DETERMINATION IN HUMAN BEINGS

1. Chromosomes in human beings
- Human body cells contain 23 pairs of chromosomes.
- Out of these, 22 pairs are called autosomes.
- The remaining 1 pair is called sex chromosomes.
- Both males and females have 22 pairs of autosomes.
- Females have XX sex chromosomes.
- Males have XY sex chromosomes.

2. Gametes and sex chromosomes
- Gametes (sperms and eggs) are formed by a special type of cell division called meiosis.
- During meiosis, the number of chromosomes is reduced to half.
- Each gamete in humans carries 23 chromosomes: 22 autosomes + 1 sex chromosome.

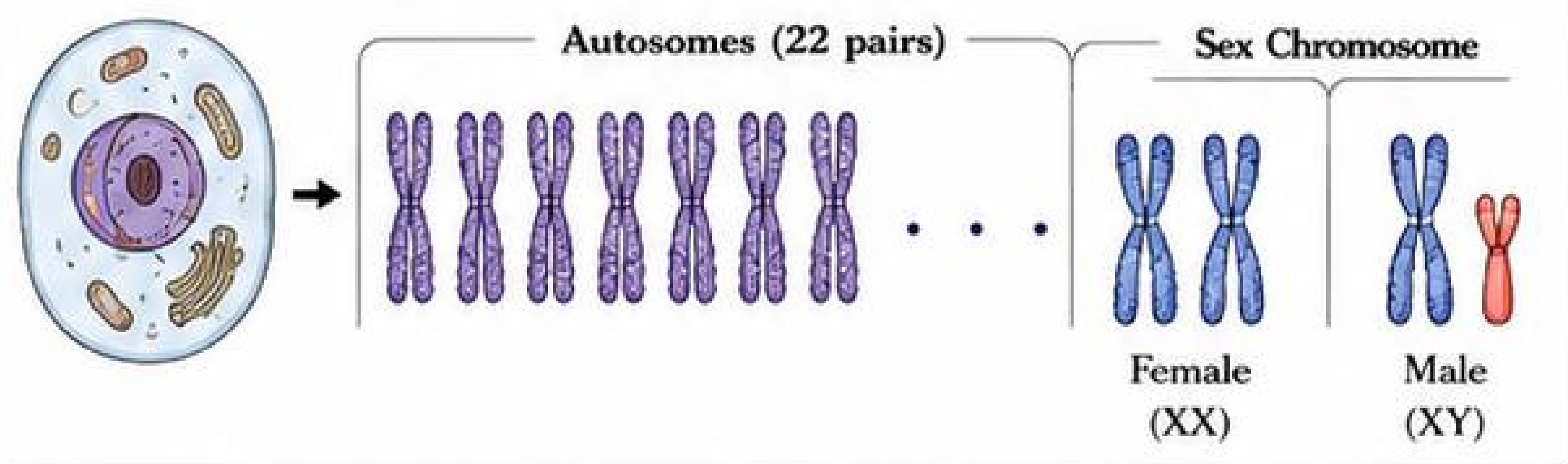
3. Gametes in females (XX)
- In females, the sex chromosome pair is XX.
- During meiosis, both X chromosomes separate.
- **All ova** (eggs) produced by a female carry only one X chromosome.
- Thus, every egg = 22 autosomes + X.

4. Gametes in males (XY)
- In males, the sex chromosome pair is XY.
- During meiosis, X and Y chromosomes separate.
- Half of the sperms carry X chromosome.
- The other half carry Y chromosome.
- Thus, two types of sperms are produced: 22 autosomes + X and 22 autosomes + Y.

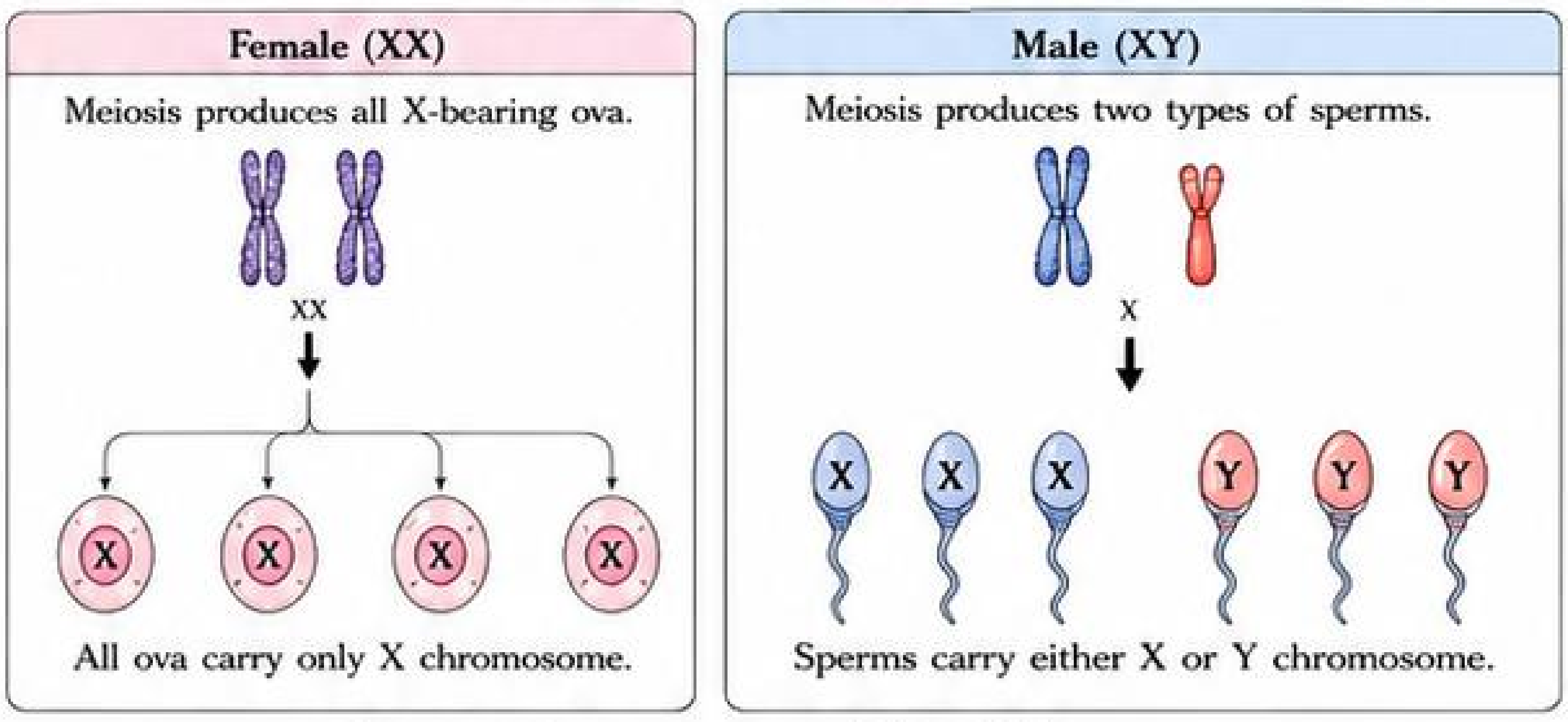
5. Fertilisation and sex determination
- At fertilisation, the egg (always X) fuses with a sperm (X or Y).
- The combination decides the sex of the child.

6. Punnett square: Cross (XX × XY)
- Mother is XX (all eggs carry X).
- Father is XY (sperms carry X or Y).
- Possible combinations and their outcome:

Chromosomes in Human Beings



Formation of Gametes



Punnett Square: Cross (XX × XY)

		Father (XY)	
		X	Y
Mother (XX) (all eggs carry X)	X	XX (Daughter)	XY (Son)
	X	XX (Daughter)	XY (Son)

Outcome
- 2 XX (Girls)
- 2 XY (Boys)
- 50% Girls
- 50% Boys

Thus, probability of girl = 50% and probability of boy = 50%.

The father determines the sex of the child.
- The egg from the mother always carries X chromosome.
- The sperm from the father decides the sex (X → **girl**, Y → boy).
- Blaming the mother for the sex of the child is scientifically incorrect and socially wrong.

Autosomes vs Sex Chromosomes

Feature	Autosomes	Sex Chromosomes
Number of pairs	22 pairs (44 chromosomes)	1 pair (2 chromosomes)
Present in	Both males and females	Determine sex
Role	Control body characters and functions	Determine sex of an individual
Same in males and females?	Yes	No (XX in females, XY in males)
Inherited from	From both parents	One from each parent
Examples of traits	Height, eye colour, blood group, skin colour, etc.	Male or female

Common Myths vs Facts

Myth	Fact
Mother decides the sex of the child.	False. The mother always gives X. The father's sperm (X or Y) decides the sex.
A mother can control whether she conceives a boy or a girl.	False. The type of sperm that fertilises the egg is random and cannot be controlled by the mother.
If a family has only girls, there is something wrong with the mother.	False and unjust. It depends on the father's sperms carrying only X in those cases.
Y chromosome is stronger than X.	False. X and Y are different, not stronger/weaker. They simply determine sex.

KEY IDEA

Humans have 23 pairs of chromosomes: 22 pairs of autosomes and 1 pair of sex chromosomes (XX in females, XY in males). The father determines the sex of the child.

EXAM TIPS – CENTUM FOCUS

1. Learn the human chromosome number: 23 pairs = 46.
2. Remember: XX = female, XY = male.
3. Egg always carries X; sperm carries X or Y.
4. Punnett square for XX × XY → 50% XX (girl), 50% XY (boy).
5. In exams, write clearly: "Sex is determined by the father."

QUICK RECALL
- **23 pairs:** 22 autosomes + 1 sex pair
- Femaler (XX) → all eggs carry X
- Male (XY) → sperms carry X or Y
- XX child = Girl, XY child = Boy
- **Probabilities:** 50% Girl, 50% Boy
- Father determines the sex

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SECTION H - KEY TERMS, COMPARISONS AND HIGH-SCORE TABLES

CORE COMPARISONS AT A GLANCE

Comparison	Term 1	Term 2	Key Difference (Why it matters)
Heredity vs Variation	• Transmission of traits from parents to offspring. • Ensures continuity of species. • Based on genetic similarity.	• Differences in traits among individuals of same species. • Creates diversity and new combinations. • Caused by genes & environment.	Heredity keeps traits similar across generations; variation creates differences that help species adapt and evolve.
Gene vs Allele	• Gene is a unit of heredity. • Located on chromosomes. • Determines a trait.	• Alleles are different forms of the same gene. • Located at the same locus.	Gene = blueprint; Alleles = different versions of that blueprint.
DNA vs Chromosome	• DNA is a molecule. • Double helix of nucleotides (A, T, G, C). • Carries genetic information.	• Chromosome is a structure. • Made of long DNA molecules wound around proteins (histones). • Found in the nucleus.	DNA is the material; chromosome is the packaged form that fits in the nucleus.
Dominant vs Recessive	• Dominant allele expresses in the presence of a recessive allele. • Represented by a capital letter (e.g., T).	• Recessive allele expresses only when no dominant allele is present. • Represented by a small letter (e.g., t).	Dominant shows in hybrid condition; recessive is masked by dominant.
Genotype vs Phenotype	• Genotype is the genetic makeup. • Combination of alleles a person carries (e.g., TT, Tt, tt). • Not visible.	• Phenotype is the observable trait. • Physical expression of genotype (e.g., tall, dwarf). • Visible.	Genotype = what you have; Phenotype = what you show.
Homozygous vs Heterozygous	• Homozygous: Same alleles for a gene (TT or tt). • Pure for that trait.	• Heterozygous: Different alleles for a gene (Tt). • Hybrid for that trait.	Homozygous = identical alleles; Heterozygous = different alleles.
Inherited Traits vs Acquired Traits	• Inherited from parents through genes. • Present from birth. • Cannot be changed.	• Developed during life due to environment, practice, habit. • Not passed to offspring.	Inherited are genetic and heritable; acquired are learned and non-heritable.
Autosomes vs Sex Chromosomes	• Non-sex chromosomes. • Same in males and females (22 pairs in humans). • Carry genes for body traits.	• Sex chromosomes determine sex. • XX in females, XY in males. • Carry genes for sex-linked traits.	Autosomes = body traits; Sex chromosomes = sex determination & sex-linked traits.

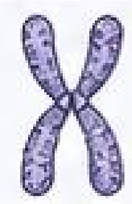
FLOW OF HEREDITY: MINI CONCEPT MAP

Cell
Basic unit of life. Contains a nucleus.

↓

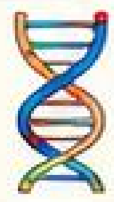
Nucleus
Control centre of the cell. Contains chromosomes.

↓



Chromosome
Thread-like structures made of DNA and proteins. Carry genes.

↓



DNA
Double helix molecule carrying genetic information.

↓

Gene
A specific segment of DNA that codes for a trait.

↓

Trait
Observable characteristic (e.g., eye colour, hair type).

↓

Inheritance
Transmission of traits from parents to offspring.

↓

Variation
Differences in traits among individuals of the same species.

HOW TO SOLVE PUNNETT SQUARE QUESTIONS (ALGORITHM)

1. Identify the trait and write the symbols for alleles. (Use capital for dominant, small for recessive).
2. Write the genotype of both parents.
3. Find the possible gametes each parent can produce.
4. Draw a Punnett square: Write one parent's gametes across the top and the other's down the side.
5. Fill in the boxes by combining alleles.
6. Write the genotypes of all offspring in the boxes.
7. Count and write the genotypic ratio.
8. Write the phenotypic ratio using dominance rule.

Example: T (Tall) dominant over t (Dwarf)
Cross: Tt × Tt

	T	t
T	TT	Tt
t	Tt	tt

Genotypic ratio: 1 TT : 2 Tt : 1 tt

Phenotypic ratio: 3 Tall : 1 Dwarf

Tip: Always write gametes first, then combine carefully.

ONE-LINE EXAMPLES (QUICK RECALL)

- Tall vs dwarf plant: Tall (T) dominant over dwarf (t).
- Round vs wrinkled pea seeds: Round (R) dominant.
- Tongue rolling: Roll (R) dominant, Non-roll (r) recessive.
- Widow's peak: Present (W) dominant, Absent (w) recessive.
- Attached vs free earlobes: Attached (A) dominant.
- Blood groups (ABO): I^A, I^B, i (multiple alleles).
- Human sex: XX (female), XY (male).
- Colour blindness: X-linked recessive trait.

MEMORY TIPS (EASY TO REMEMBER)

- "Heredity = Handover" (traits passed on)
- "Variation = Variety" (difference creates diversity)
- Gene makes; Allele changes.
- DNA = Data; Chromosome = Chassis.
- "Big D shows up" (Dominant expresses).
- "Recessive rests" (only appears without dominant).
- Genotype in books; Phenotype in looks.
- Homo = Same; Hetero = Different.
- Inherited in DNA; Acquired in Daily Life.

WARNING SIGNS (COMMON MISTAKES)

1. Writing same allele combination in all boxes.
2. Forgetting to use gametes, writing parents' genotype directly.
3. Confusing phenotype with genotype.
4. Ignoring dominance while writing phenotype.
5. Reversing ratios (genotypic vs phenotypic).
6. Forgetting that acquired traits are not heredity.

HIGH-YIELD EXAM POINTERS

- ✓ Understand the levels: DNA → Gene → Trait.
- ✓ Always check dominance before writing phenotype.
- ✓ Punnett squares: neat layout = more marks.
- ✓ Learn key examples and ratios (3:1, 1:2:1).
- ✓ Diagrams (cell, chromosome, DNA) fetch marks.
- ✓ Write definitions in your own words first.
- ✓ Use keywords: gene, allele, chromosome, trait, heredity, variation, environment.

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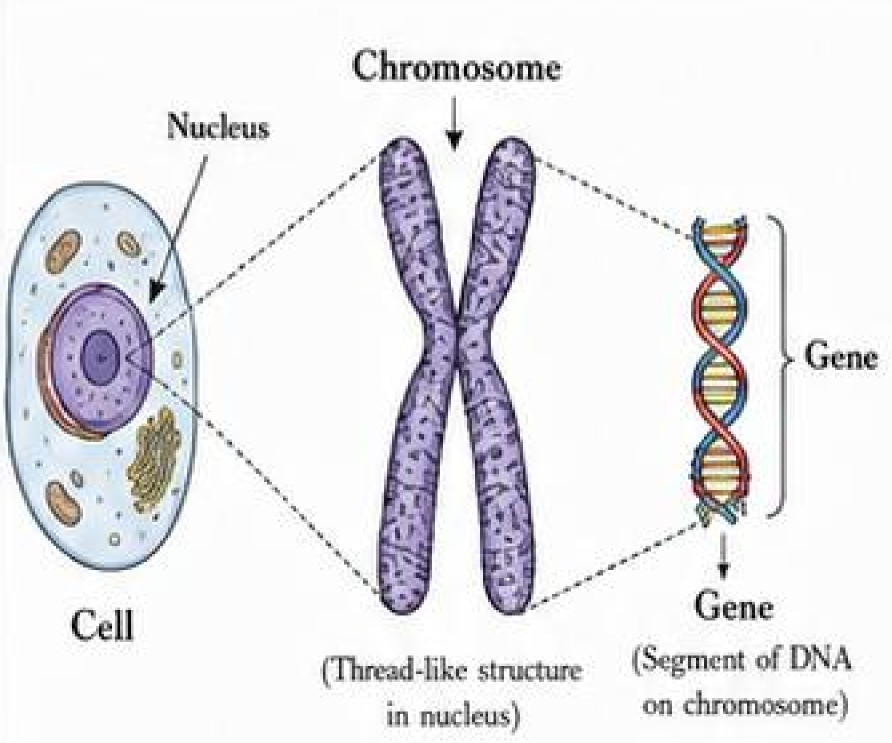
SECTION I - IMPORTANT DIAGRAMS, QUESTION PRACTICE AND ANSWER FRAMES

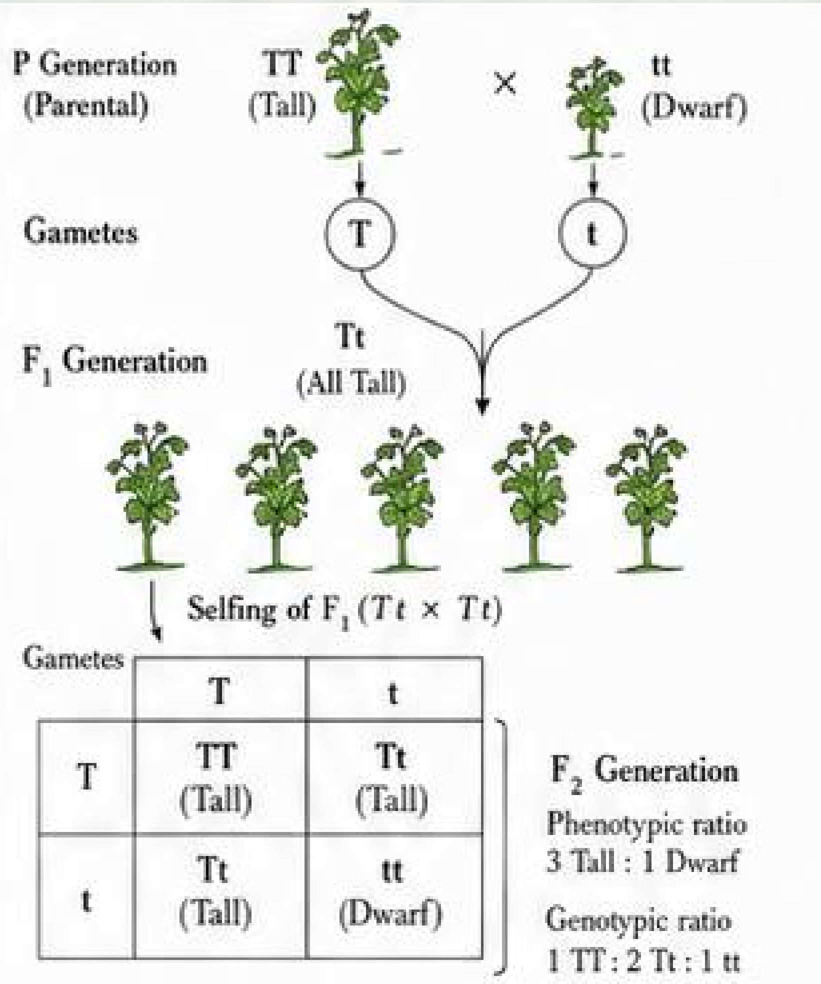
1. Cell → Chromosome → Gene Relationship



Key Points to Write

- A cell contains a nucleus.
- Chromosomes are thread-like structures present in the nucleus.
- Chromosomes are made of DNA.
- A gene is a specific segment of DNA on a chromosome.
- Genes control traits and are passed from parents to offspring.

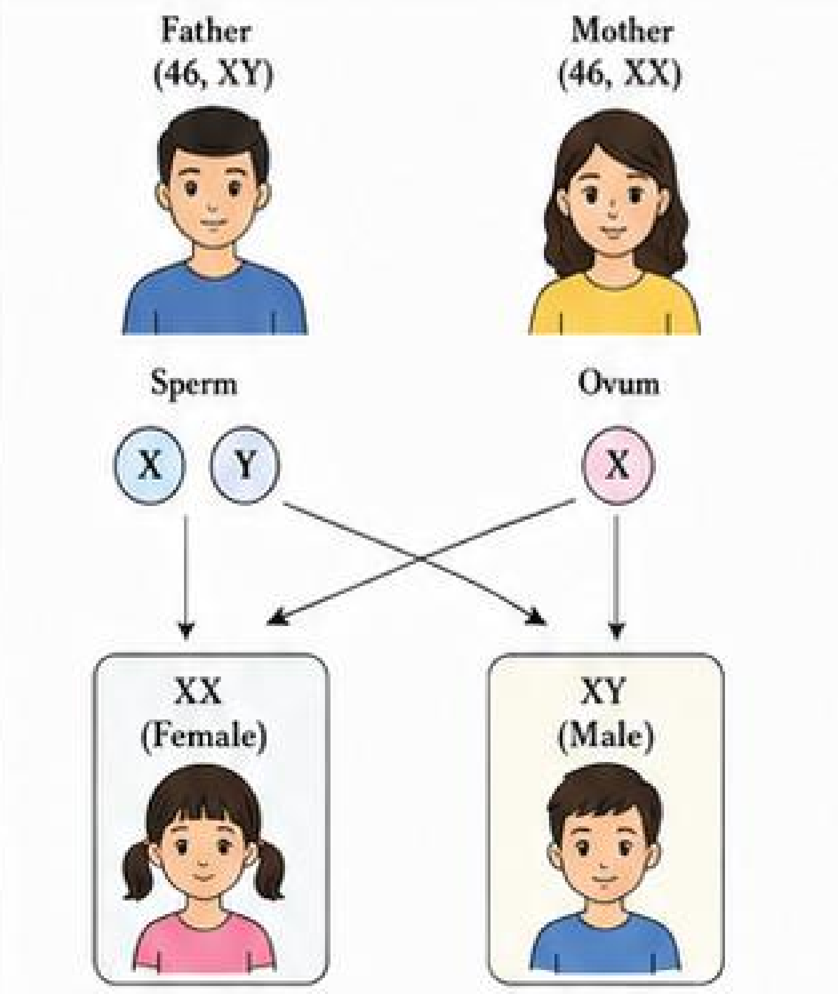
2. Mendel's Monohybrid Cross (Overview)



Key Points to Write

- Mendel studied one pair of contrasting traits at a time.
- Cross between pure tall (TT) and pure dwarf (tt) pea plants.
- F_1 generation: all tall (Tt).
- Selfing of F_1 gives F_2 with 3 tall : 1 dwarf.
- Genotypic ratio is 1 TT : 2 Tt : 1 tt.

3. Sex Determination in Humans



Probability: 50% Female (XX) : 50% Male (XY)

Key Points to Write

- Humans have 23 pairs of chromosomes.
- Females are XX and males are XY.
- Mother produces only X-bearing ova.
- Father produces X-bearing and Y-bearing sperms.
- XX result in female; XY result in male.

Answer Frames for Common Exam Questions

1. Define heredity.

Answer Frame:

Heredity is the ________ of characteristics (traits) from ________ to their ________ through genes.

2. Define variation.

Answer Frame:

Variation refers to the ________ in characteristics among individuals of the same ________.

3. Explain Mendel's monohybrid experiment.

Answer Frame:

Mendel crossed pure ________ (TT) and pure ________ (tt) pea plants. All F_1 were ________ (Tt). On selfing F_1, F_2 showed ________ : ________ phenotypic ratio (3:1) and genotypic ratio 1 : 2 : 1.

4. Distinguish between inherited and acquired traits.

Answer Frame:

Inherited traits are ________ from parents through ________, present from ________ and not influenced by environment. Acquired traits develop during life due to ________, ________, and are not passed to offspring.

5. Explain sex determination in humans.

Answer Frame:

Females are ________ (XX) and males are ________ (XY). Mother produces ________ ova. Father produces ________ and ________ sperms. If sperm with ________ fertilizes ova, offspring is female (XX). If sperm with ________ fertilizes ova, offspring is male (XY).

2-MARK QUESTIONS (SHORT ANSWER TYPE)

1. What is heredity?
   Answer: Heredity is the transmission of characteristics (traits) from parents to their offspring through genes.
2. What is variation?
   Answer: Variation refers to the differences in characteristics among individuals of the same species.
3. Name the scientist who is known as the 'Father of Genetics'.
   Answer: Gregor Johann Mendel.

3-MARK QUESTIONS (SHORT ANSWER TYPE)

1. State the law of segregation.
   Answer: During the formation of gametes, the two alleles of a gene separate from each other so that each gamete gets only one allele. These alleles reunite during fertilisation.
2. Why do offspring show resemblance to their parents?
   Answer: Offspring receive genes from both parents. These genes control the traits and are the same as those present in parents, so they show resemblance to them.
3. Write the phenotypic and genotypic ratio of a monohybrid cross.
   Answer: Phenotypic ratio – 3 : 1 (Tall : Dwarf)
   Genotypic ratio – 1 : 2 : 1 (TT : Tt : tt)

WRITE THESE KEYWORDS FOR FULL MARKS

- Heredity • Variation • Trait • Gene • Chromosome • DNA • Inherited Traits • Acquired Traits
- Mendel • Monohybrid Cross • Dominant • Recessive • Genotype • Phenotype • Law of Segregation
- Gametes • Fertilisation • XX • XY • Male • Female • Offspring • Environment • Adaptation • Survival

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SECTION J - MASTER REVISION SHEET, NCERT TRAPS AND CENTUM FOCUS

1. Compact Definitions – Key Terms

Term	Definition
Heredity	Transmission of characteristics (traits) from parents to their offspring.
Variation	Differences in characteristics among individuals of the same species.
Trait	A specific characteristic, e.g., tall, round seeds, blood group.
Gene	A segment of DNA that controls a particular trait.
Allele	Different forms of the same gene.
Dominant allele	An allele expressed in the presence of another allele.
Recessive allele	An allele expressed only in the absence of a dominant allele.
Genotype	Genetic makeup (combination of alleles) of an individual.
Phenotype	Observable physical expression of a trait.
Homozygous	Having two identical alleles for a trait (e.g., TT or tt).
Heterozygous	Having two different alleles for a trait (e.g., Tt).
Monohybrid cross	Cross between individuals differing in one pair of traits.
Dihybrid cross	Cross between individuals differing in two pairs of traits.
True-breeding	Homozygous individuals that produce same offspring for a trait.
Punnett square	A diagram used to predict possible genotype/phenotype of offspring.
Law of Dominance	In a pair of contrasting traits, one masks the expression of the other.
Law of Segregation	The two alleles of a gene separate during gamete formation.
Law of Independent Assortment	Alleles of different genes assort independently during gamete formation.

2. Master Summary – Mendel's Results

1. Mendel selected garden pea because it has clear contrasting traits, flowers are bisexual, natural self-pollination occurs, many seeds are produced, and many varieties are available.
2. He performed hybridisation and selfing in a planned way and analysed the results.
3. In monohybrid cross, F_1 generation showed only dominant trait.
4. On selfing F_1, F_2 generation showed both dominant and recessive traits in a 3:1 ratio.
5. On test cross of F_1 (Tt × tt), offspring showed 1:1 ratio of dominant : recessive.
6. In dihybrid cross, F_2 generation showed 9:3:3:1 phenotypic ratio.
7. Mendel concluded:

- Traits are controlled by discrete units (genes).
- Alleles segregate during gamete formation (*Law of Segregation*).
- Alleles of different genes assort independently (*Law of Independent Assortment*).

8. These results form the basis of modern genetics.

3. Genotype and Phenotype Ratios – At a Glance

Cross Type	Cross	Genotypic Ratio (F_2)	Phenotypic Ratio (F_2)
Monohybrid	Tt × Tt	1 TT : 2 Tt : 1 tt	3 Tall : 1 Dwarf
Test Cross	Tt × tt	1 Tt : 1 tt	1 Tall : 1 Dwarf
Dihybrid	TtYy × TtYy	1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1 (Total 9 genotypes)	9 : 3 : 3 : 1 (Round Yellow : Round Green : Wrinkled Yellow : Wrinkled Green)

4. Do-Not-Forget – Key Points

★ Genes are located on chromosomes.
★ Alleles of a gene separate during gamete formation.
★ Dominant allele masks the expression of recessive allele.
★ Traits are inherited in a predictable manner.
★ Environment can influence the expression of traits.
★ Mutation creates new genes/alleles.
★ Sex determination in humans: XX (female), XY (male).
★ Blood group is an example of multiple alleles.
★ Variation is essential for survival and evolution.

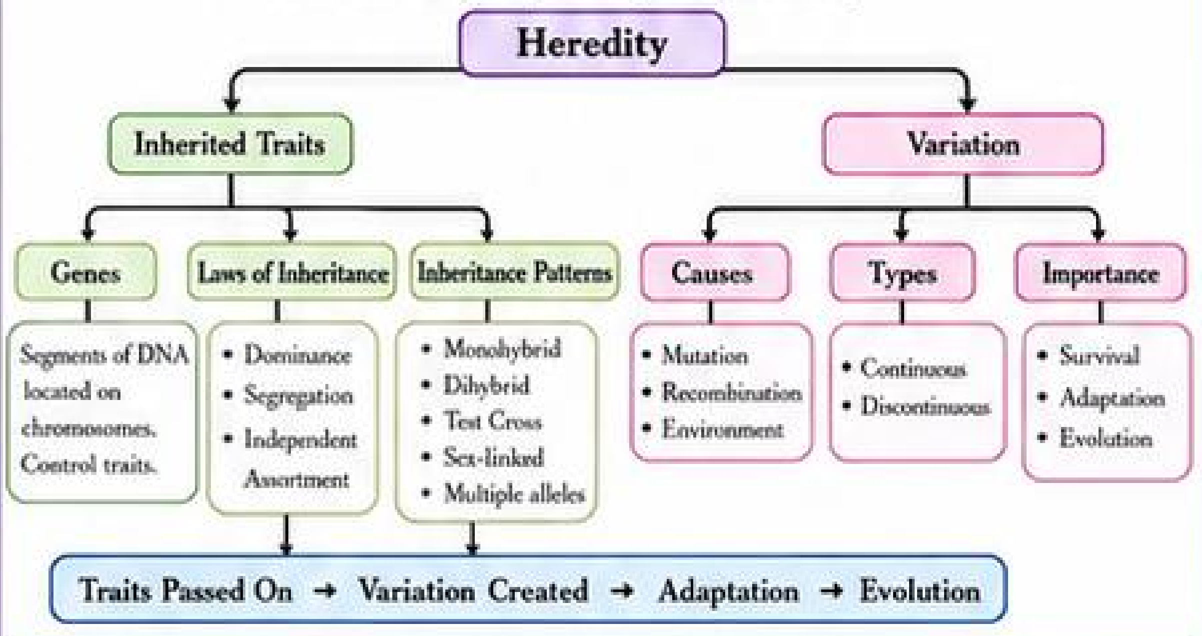
5. NCERT Traps – Common Misconceptions & Correct Explanations

Misconception / Trap	Correct Explanation
1. F_1 generation shows both dominant and recessive traits.	F_1 shows only dominant trait. Recessive trait appears in F_2.
2. In Tt × Tt cross, all tall plants are TT.	Tall plants can be TT or Tt. Only TT is homozygous tall.
3. Law of Segregation applies to different genes.	It applies to alleles of the same gene.
4. Traits are blended in offspring.	No blending in inheritance. Traits remain discrete.
5. Environment does not affect traits.	Environment affects expression of traits, not genes.
6. Blood group is controlled by one gene with two alleles.	Blood group is controlled by one gene with three alleles (I^A, I^B, i).
7. Y-linked traits are passed from mother to son.	Y-linked traits are passed from father to son.
8. Identical twins have different DNA.	Identical twins have same DNA; differences are due to environment.

6. Compare and Contrast

Feature	Monohybrid Cross	Dihybrid Cross
Number of trait pairs	One	Two
F_1 generation	All show dominant trait	All show dominant traits
F_2 phenotypic ratio	3 : 1	9 : 3 : 3 : 1
F_2 genotypic ratio	1 : 2 : 1	1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1
Law involved	Segregation	Segregation + Independent Assortment
Punnett square size	2 × 2	4 × 4
Example	Height of pea plant	Seed shape and seed colour

7. Final Concept Map – Heredity



8. Super Quick Recall – 10 Must Remember

1. Gene is a segment of DNA.
2. Alleles of a gene separate (Law of Segregation).
3. Dominant allele masks recessive allele.
4. Monohybrid cross → 3:1 phenotypic ratio.
5. Test cross of Tt → 1:1 ratio.
6. Dihybrid cross → 9:3:3:1 phenotypic ratio.
7. Sex in humans: XX female, XY male.
8. Blood group has three alleles: I^A, I^B, i.
9. Variation helps in survival and evolution.
10. Environment affects expression, not genes.

9. Centum Focus – Revise This Before Exam

✓ Key terms and definitions
✓ Mendel's pea plant experiment
✓ Laws of Inheritance (3 Laws)
✓ Monohybrid cross – steps and ratios
✓ Dihybrid cross – steps and ratios
✓ Test cross – concept and result
✓ Sex determination in humans
✓ Blood groups and multiple alleles
✓ Sources and types of variation
✓ Importance of variation
✓ Diagrams – nucleus, chromosome, Punnett square
✓ NCERT Examples & In-Text Questions
✓ All back exercise short & long answers

Revise smart, label diagrams neatly, and score full marks.